

Carry-Free Radix Operators:

Exact Schatten Surfaces, Binary Endpoint Pinching, and Weighted Cycle Traces

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Abstract

For an integer radix $b \geq 2$, we study the infinite matrix on the positive integers whose (m, n) entry is $\mathbf{1}_{\{m+n \text{ is carry-free in base } b\}}(mn)^{-s/2}$. Writing $\sigma = \Re s$, we prove for every $1 \leq q < \infty$ the exact classification

$$B_{b,s} \in S_q \iff \sigma > \max\{1, \log_b \|C_b\|_{S_q}\},$$

where $C_b = (\mathbf{1}_{a+c < b})_{0 \leq a, c < b}$. Its singular values are $[2 \sin((2j-1)\pi/(4b+2))]^{-1}$, so the critical surface is explicit. The proof separates a universal column wall from a digit-norm wall through exact radix-shell factorizations. At $b = 2$, every same-shell block vanishes; paired adjacent-shell compressions recover the full nonmembership range, including equality. Consequently boundedness, compactness, and Hilbert–Schmidt membership are all equivalent to $\sigma > 1$, whereas trace class begins at $\sigma > \alpha_b = \log_b \|C_b\|_{S_1} > 1$. We also identify the legal domains of the trace, trace powers, the regularized determinant, and the least-period support of the associated countable graph shift. Independent finite lanes, interval checks, and hostile mutations provide consistency evidence for the exact identities; they are not used as proofs of the infinite statements.

1 Introduction

Digitwise constraints are local, but the operators they generate need not have local analytic behavior. A simple example is the rule that two nonnegative base- b digits may be adjacent exactly when their sum is smaller than b . As a zero-completed fixed-length control, this rule gives a finite Kronecker power. On the positive integers, however, word length is unbounded, the all-zero word is absent, and the Dirichlet weight interacts with both the number and the length of the admissible words; the finite control is not a finite restriction of this positive-vertex source. The resulting infinite operator therefore has two different sources of divergence.

Write the base- b expansions as

$$m = \sum_{j \geq 0} m_j b^j, \quad n = \sum_{j \geq 0} n_j b^j, \quad 0 \leq m_j, n_j < b,$$

with only finitely many nonzero digits. We say that $m + n$ is *carry-free in base b* , and write $m \perp_b n$, if $m_j + n_j < b$ for every j . For $s \in \mathbb{C}$, consider the formal matrix

$$B_{b,s}(m, n) = \mathbf{1}_{\{m \perp_b n\}}(mn)^{-s/2}, \quad m, n \in \mathbb{N} = \{1, 2, \dots\}, \quad (1)$$

where complex powers use the real logarithm. Whenever this matrix has a bounded completion, we use the same symbol for the resulting operator on $\ell^2(\mathbb{N})$.

The finite digit matrix

$$C_b = (\mathbf{1}_{\{a+c < b\}})_{0 \leq a, c < b}$$

contains the local rule. Set

$$\kappa_{b,q} = \|C_b\|_{S_q}, \quad \tau_b = \kappa_{b,1}, \quad \alpha_b = \log_b \tau_b.$$

Our central result is the following exact surface.

Theorem 1.1 (Exact Schatten surface). *For every integer $b \geq 2$, every $1 \leq q < \infty$, and every $s \in \mathbb{C}$ with $\sigma = \Re s$,*

$$B_{b,s} \in S_q \iff \sigma > \max\{1, \log_b \kappa_{b,q}\}. \quad (2)$$

Moreover, boundedness, compactness, and Hilbert–Schmidt membership are each equivalent to $\sigma > 1$, while trace-class membership is equivalent to $\sigma > \alpha_b$, where $\alpha_b > 1$.

The maximum in equation (2) is structural. The first wall, $\sigma = 1$, is already visible in one positive-density column. The second comes from the exponential S_q -growth of the digit tensors. It is an active endpoint only when $\log_b \kappa_{b,q} > 1$; otherwise the universal wall hides it. Equality at an active digit wall is nevertheless delicate. For $b \geq 3$, diagonal radix shells provide a direct orthogonal pinching. For $b = 2$, every such diagonal block is exactly zero, and equality must be recovered from disjoint pairs of adjacent shells.

The digit constants are explicit:

$$s_j(C_b) = (2 \sin \frac{(2j-1)\pi}{4b+2})^{-1}, \quad 1 \leq j \leq b. \quad (3)$$

For example, $\tau_2 = \sqrt{5}$ and $\alpha_2 = \log_2 \sqrt{5}$, whereas $\kappa_{b,2}^2 = b(b+1)/2 < b^2$. Thus the trace-class wall lies strictly to the right of 1, but the Hilbert–Schmidt digit wall lies strictly to its left for every radix.

Beyond ideal membership, the same domain bookkeeping determines which cycle expressions are legal. In $\sigma > 1$, all powers $B_{b,s}^r$, $r \geq 2$, are trace class and their traces are absolutely convergent weighted closed-walk sums. Consequently $\det_2(I - zB_{b,s})$ is entire in z , with its trace-power logarithm only asserted locally near $z = 0$. The ordinary trace and Fredholm determinant require the smaller trace-class domain $\sigma > \alpha_b$. The positive-vertex convention is essential here: the all-zero digit word must be removed before traces or temporal support are interpreted.

The mechanism of the proof is summarized in figure 1. It also shows the evidence boundary used throughout the paper: finite tensors and computations check identities, but the infinite classification comes from analytic shell summation and endpoint pinching.

Contributions and scope. The paper establishes three linked statements.

1. It gives a self-contained classification of $B_{b,s}$ in every Banach Schatten ideal S_q , for every integer radix, including both strict walls.
2. It separates the higher-radix same-shell endpoint mechanism from the binary paired-shell mechanism and derives the exact bounded, compact, Hilbert–Schmidt, and trace-class corollaries.
3. It records the legal weighted trace and determinant domains and the least-period support of the countable carry-free graph shift, with the zero word deleted.

These statements do not assign new ownership to classical carry criteria, finite Pascal or Boolean matrices, finite disjointness tensors, or the general theory of trace ideals. Kummer’s and Lucas’s classical prime-modular results provide historical arithmetic context [5, 8]; standard ideal and regularized-determinant terminology follows Simon [9]. The all-radix relation used here is defined directly, so no composite-radix version of Kummer’s theorem is invoked. A bounded directed search did not find the same-quantifier infinite weighted theorem, but that absence is not a priority or exhaustiveness claim.

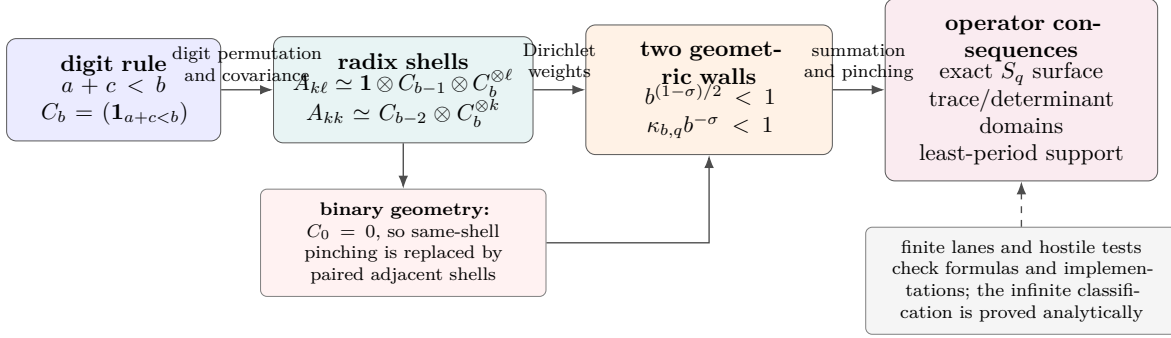


Figure 1: The proof pipeline. Exact digit singular values determine unweighted shell norms; Dirichlet weights turn them into a universal shell-separation ratio and a digit-growth ratio. Their strict summability, together with the appropriate endpoint pinching, yields the ideal surface and its trace, determinant, and period consequences. The lower boxes isolate the binary zero-block exception and the proof-versus-validation boundary.

Organization. Section 2 fixes the ownership boundary. Section 3 derives the digit spectrum, and section 4 turns it into exact shell formulas. Section 5 proves the sharp surface and both endpoint mechanisms. Section 6 treats traces, determinants, and least periods. Section 7 reports finite consistency controls without using them as proof. The appendices expand the digit recurrence, shell coordinates, endpoint limiting arguments, and reproducibility ledger.

2 Related work and ownership boundary

Carries and congruences. For a prime radix, carry counts are classically tied to prime-adic valuations of binomial coefficients through Kummer’s theorem [5]; Lucas’s congruence gives a complementary digitwise description modulo a prime [8]. Those results own the relevant arithmetic precedents. Our source relation, however, is simply the coordinatewise inequality $m_j + n_j < b$, defined for every integer $b \geq 2$. We use the classical theorems only as prime-radix comparators and never as a definition or proof in composite radix.

Finite Pascal, Boolean, and disjointness matrices. Finite graphs built from binomial parity and their spectra were studied by Christopher and Kennedy [4]; symmetric Pascal matrices modulo a prime have a substantial finite theory [2]. LaGrange treats eigenvalues of Boolean graphs and Pascal-type matrices [6]. Related disjointness matrices also occur in communication-complexity factorization norms [7], rectifier-network coverings [3], and algorithms for Kronecker powers [1]. These works establish ownership of the finite arithmetic, graph, tensor, spectral, covering, and algorithmic mechanisms that overlap our controls. Accordingly, we claim no novelty for finite digit Kronecker products, finite carry tests, finite singular values as objects, or finite Boolean/disjointness spectra.

Trace ideals and determinants. We use the standard Schatten ideals and the usual Hilbert–Schmidt regularized determinant $\det_2$; their general theory is described, for example, by Simon [9]. All estimates particular to the carry-free operator—the shell factorizations, the uniform transfer from unweighted to weighted blocks, the two endpoint pinchings, and the closed-walk limiting argument—are supplied below.

Positioning. The distinction relevant to this paper is between a fixed finite digit tensor and an infinite positive-vertex operator with Dirichlet weights and unbounded digit length. The

latter requires simultaneous control of shell separation, digit growth, endpoint equality, and deletion of the all-zero word. The bounded source search performed before writing did not locate the exact combination of quantifiers in theorem 1.1; this statement reports the search outcome only and does not establish priority.

3 The carry-free source and the digit operator

3.1 Conventions and phase removal

For $d \geq 1$, define

$$C_d = (\mathbf{1}_{\{a+c < d\}})_{0 \leq a, c < d}, \quad \kappa_{d,q} = \|C_d\|_{S_q},$$

and let C_0 denote the zero block, with $\kappa_{0,q} = 0$. The positive-vertex alphabet is always $\mathbb{N} = \{1, 2, \dots\}$. When a length- L tensor $C_b^{\otimes L}$ is used as a finite control, its all-zero word is therefore not a vertex of the infinite source.

The imaginary part of s has no effect on singular values, but the relation is a two-sided unitary equivalence rather than a unitary conjugation. Indeed, for $s = \sigma + it$, let

$$U_t e_n = n^{-it/2} e_n.$$

Entrywise,

$$B_{b,s} = U_t B_{b,\sigma} U_t. \quad (4)$$

Both factors are unitary, so every singular-value assertion can be proved for the real symmetric matrix $B_{b,\sigma}$. We do not infer from equation (4) that $B_{b,s}$ is positive or self-adjoint when $t \neq 0$.

3.2 Exact singular values

Proposition 3.1 (Digit spectrum). *The singular values of C_b , in decreasing order, are*

$$s_j(C_b) = [2 \sin \frac{(2j-1)\pi}{4b+2}]^{-1}, \quad 1 \leq j \leq b. \quad (5)$$

Consequently,

$$\kappa_{b,q} = [\sum_{j=1}^b (2 \sin \frac{(2j-1)\pi}{4b+2})^{-q}]^{1/q}. \quad (6)$$

Proof. The (a, a') entry of $C_b C_b^*$ counts the digits c satisfying $c < b - \max(a, a')$, and hence equals $b - \max(a, a')$. Reverse the row and column order and index the result by $1, \dots, b$. The resulting matrix is

$$K_b = (\min(i, j))_{1 \leq i, j \leq b}.$$

Its inverse is the tridiagonal matrix

$$L_b = K_b^{-1} = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 1 \end{pmatrix}.$$

For completeness, multiplying K_b by this matrix subtracts consecutive columns twice in the interior and once at the terminal boundary, producing the identity.

b	τ_b	α_b	$\kappa_{b,2}$	$\log_b \kappa_{b,2}$	$\sigma_c(q=2)$
2	2.236068	1.160964	1.732051	0.792481	1.000000
3	3.603875	1.166936	2.449490	0.815465	1.000000
4	5.064178	1.170164	3.162278	0.830482	1.000000
5	6.595934	1.172119	3.872983	0.841303	1.000000

Table 1: Display evaluations of the exact formulas in equations (6) and (7). The last column is $\sigma_c(q=2) = \max\{1, \log_b \kappa_{b,2}\}$; the decimals are not empirical estimates and do not define the theorem.

The eigenvalue equation for L_b has the interior solution $v_i = \sin(i\theta)$, with $\lambda = 2 - 2\cos\theta = 4\sin^2(\theta/2)$ and the left boundary $v_0 = 0$. The final row is equivalent to the ghost boundary $v_{b+1} = v_b$, so

$$\sin((b+1)\theta) - \sin(b\theta) = 0.$$

The b roots in $(0, \pi)$ are $\theta_j = (2j-1)\pi/(2b+1)$. Thus the eigenvalues of K_b are $[4\sin^2(\theta_j/2)]^{-1}$. Taking square roots proves equation (5), and summing their q th powers proves equation (6). $\square$

Corollary 3.2 (Two exact comparisons). *For every $b \geq 2$,*

$$\kappa_{b,2}^2 = \frac{b(b+1)}{2} < b^2, \quad \tau_b := \kappa_{b,1} > b. \quad (7)$$

In particular $\log_b \kappa_{b,2} < 1$ and $\alpha_b := \log_b \tau_b > 1$.

Proof. The Hilbert–Schmidt norm squared counts the ones in C_b , giving $b + (b-1) + \dots + 1 = b(b+1)/2$. Reversing the columns of C_b makes it triangular with unit diagonal, so $|\det C_b| = 1$. The product of its singular values is therefore one. Their arithmetic mean is at least their geometric mean, and the inequality is strict because their squared sum $b(b+1)/2$ exceeds b . Hence $\tau_b > b$. $\square$

At $b = 2$, equation (5) gives the golden-ratio pair $(1 + \sqrt{5})/2$ and $(\sqrt{5} - 1)/2$. Their sum is $\sqrt{5}$, so

$$\tau_2 = \sqrt{5}, \quad \alpha_2 = \log_2 \sqrt{5}. \quad (8)$$

These identities will distinguish the binary trace-class wall from its Hilbert–Schmidt wall.

4 Exact radix-shell calculus

Let

$$I_k = [b^k, b^{k+1}) \cap \mathbb{N}, \quad k \geq 0,$$

and let P_k be the coordinate projection from $\ell^2(\mathbb{N})$ onto $\ell^2(I_k)$. We write $A_{k\ell}$ for the unweighted carry-free matrix from I_ℓ to I_k , and

$$B_{k\ell} = P_k B_{b,\sigma} P_\ell.$$

The notation $X \stackrel{\text{sv}}{\cong} Y$ below means that X and Y have the same nonzero singular values, with multiplicity; zero rows and columns may have been deleted.

Proposition 4.1 (Exact unweighted shell blocks). *For $k > \ell$,*

$$A_{k\ell} \stackrel{\text{sv}}{\cong} \mathbf{1}_{(b-1)b^{k-\ell-1}} \otimes C_{b-1} \otimes C_b^{\otimes \ell}, \quad (9)$$

and hence

$$\|A_{k\ell}\|_{S_q} = ((b-1)b^{k-\ell-1})^{1/2} \kappa_{b-1,q} \kappa_{b,q}^\ell. \quad (10)$$

For $k = \ell$,

$$A_{kk} \stackrel{\text{sv}}{\simeq} C_{b-2} \otimes C_b^{\otimes k}, \quad \|A_{kk}\|_{S_q} = \kappa_{b-2,q} \kappa_{b,q}^k. \quad (11)$$

The case $k < \ell$ follows by transposition.

Proof. Take $m \in I_k$ and $n \in I_\ell$, with $k > \ell$. At each position $0, \dots, \ell-1$, both digits range from 0 to $b-1$, and the compatibility matrix is C_b . At position ℓ , the leading digit of n ranges from 1 to $b-1$. The row indexed by the digit $m_\ell = b-1$ is zero; after deleting it and subtracting one from the column digit of n , the compatibility matrix is C_{b-1} .

All higher digits of n vanish. The top digit m_k has $b-1$ choices, and the intervening $k-\ell-1$ digits of m have b choices each. They repeat the same lower-digit block, producing the column vector of ones in equation (9). A permutation of digit coordinates now gives the displayed tensor product.

When $k = \ell$, both leading digits lie in $\{1, \dots, b-1\}$. Subtracting one from each turns the condition $m_k + n_k < b$ into a C_{b-2} block; the lower k digit positions again give $C_b^{\otimes k}$. Tensor products multiply singular values, and the lone nonzero singular value of a length- r ones vector is $\sqrt{r}$. This proves all norm identities. $\square$

Remark 4.2 (The binary zero block). For $b = 2$, equation (11) contains C_0 , so every A_{kk} is exactly zero. On the adjacent shells, however, $\kappa_{1,q} = 1$, and equation (10) gives

$$\|A_{2j+1,2j}\|_{S_q} = \kappa_{2,q}^{2j}. \quad (12)$$

Thus the binary endpoint is not absent; it lives off the shell diagonal.

4.1 Uniform transfer to weighted blocks

Let D_k be diagonal on $\ell^2(I_k)$, with $(D_k)_{m,m} = m^{-\sigma/2}$. For real $\sigma > 0$,

$$B_{k\ell} = D_k A_{k\ell} D_\ell, \quad b^{-(k+1)\sigma/2} I \leq D_k \leq b^{-k\sigma/2} I.$$

Lemma 4.3 (Uniform weighted comparison). *For $\sigma > 0$, every $k, \ell \geq 0$, and $1 \leq q < \infty$,*

$$b^{-\sigma} b^{-(k+\ell)\sigma/2} \|A_{k\ell}\|_{S_q} \leq \|B_{k\ell}\|_{S_q} \leq b^{-(k+\ell)\sigma/2} \|A_{k\ell}\|_{S_q}. \quad (13)$$

Proof. The upper bound is the ideal inequality $\|RXT\|_{S_q} \leq \|R\| \|X\|_{S_q} \|T\|$. For the lower bound, write $A_{k\ell} = D_k^{-1} B_{k\ell} D_\ell^{-1}$ and use $\|D_k^{-1}\| \leq b^{(k+1)\sigma/2}$ and the analogous estimate for D_ℓ^{-1} . The resulting constants are independent of the shell indices, which is what preserves equality endpoints. $\square$

Combining equations (10) and (13) and putting $k = \ell + h$, $h \geq 1$, yields

$$\|B_{\ell+h,\ell}\|_{S_q} \leq c_{b,q} b^{h(1-\sigma)/2} (\kappa_{b,q} b^{-\sigma})^\ell, \quad c_{b,q} = \sqrt{\frac{b-1}{b}} \kappa_{b-1,q}. \quad (14)$$

The two ratios in equation (14) are the two walls of the theorem. The exact critical curves obtained from equation (6) are shown in figure 2.

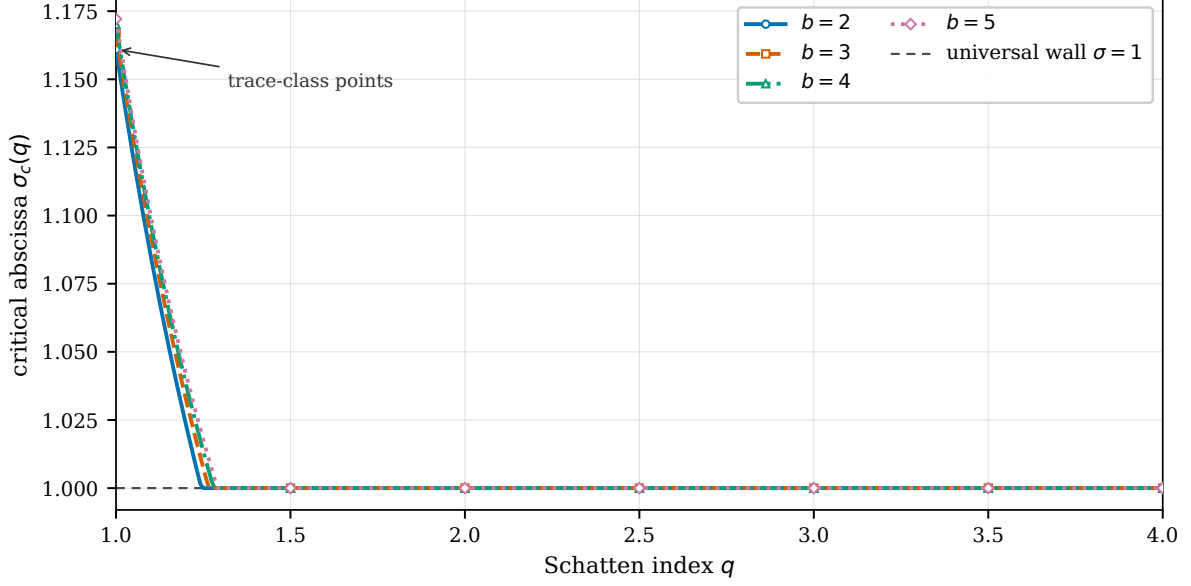


Figure 2: The exact critical abscissa $\sigma_c(q) = \max\{1, \log_b \kappa_{b,q}\}$, evaluated from equation (6) for four radices. The points at $q = 1$ are the trace-class thresholds. Once the digit curve falls below 1, the universal column wall is active and produces the plateau. Line styles and markers duplicate the color encoding.

5 Sharp ideal thresholds

5.1 Sufficiency

Proposition 5.1 (Summable side of the surface). *If*

$$\sigma > 1 \quad \text{and} \quad \kappa_{b,q} b^{-\sigma} < 1, \quad (15)$$

then $B_{b,s} \in S_q$.

Proof. By the two-sided unitary relation equation (4), it is enough to take $s = \sigma$. Sum the embedded shell blocks first over a finite square $0 \leq k, \ell \leq K$. For the strict upper triangle, put $k = \ell + h$. Equation (14) gives

$$\sum_{\ell \geq 0} \sum_{h \geq 1} \|B_{\ell+h,\ell}\|_{S_q} \leq c_{b,q} \sum_{h \geq 1} b^{h(1-\sigma)/2} \sum_{\ell \geq 0} (\kappa_{b,q} b^{-\sigma})^\ell.$$

Both geometric series converge under equation (15). The lower triangle has the same block norms. On the diagonal, equations (11) and (13) gives

$$\sum_{k \geq 0} \|B_{kk}\|_{S_q} \leq \kappa_{b-2,q} \sum_{k \geq 0} (\kappa_{b,q} b^{-\sigma})^k.$$

Because $q \geq 1$, S_q is a Banach ideal. The finite block sums are therefore Cauchy in S_q , and their limit has the prescribed matrix entries. This constructs the completed operator and proves the claim. $\square$

5.2 The universal wall

Proposition 5.2 (Column obstruction). *If $\sigma \leq 1$, the formal matrix in equation (1) has no bounded completion on $\ell^2(\mathbb{N})$.*

Proof. The column indexed by $n = 1$ contains every m whose units digit lies in $\{0, \dots, b-2\}$, because adding 1 creates no carry exactly on that residue set. The squared norm of this column is therefore at least

$$\sum_{\substack{m \geq 1 \\ m \bmod b \in \{0, \dots, b-2\}}} m^{-\sigma}.$$

For $0 < \sigma \leq 1$, this is a finite union of arithmetic-progression subseries of the divergent p -series; for $\sigma \leq 0$, its summands do not tend to zero. Hence the column is not in ℓ^2 , which is incompatible with boundedness. $\square$

The universal obstruction has no digit-spectrum input. It remains the active wall whenever $\log_b \kappa_{b,q} \leq 1$; in that case the point $\sigma = \log_b \kappa_{b,q}$ is already on the unbounded side and is not a second analytic endpoint.

5.3 The higher-radix digit wall

We use the standard orthogonal block pinching: if (Q_j) are mutually orthogonal projections, then

$$\Phi(T) = \bigoplus_j Q_j T Q_j$$

is contractive on S_q , $1 \leq q < \infty$, and

$$\|\Phi(T)\|_{S_q}^q = \sum_j \|Q_j T Q_j\|_{S_q}^q. \quad (16)$$

The statement follows first for finite families by averaging over diagonal unitaries and then for countable families by monotone truncation.

Suppose $\sigma > 1$ and

$$\rho_{b,q}(\sigma) := \kappa_{b,q} b^{-\sigma} \geq 1.$$

For $b \geq 3$, pinch $B_{b,\sigma}$ with the shell projections P_k . The lower half of equation (13) gives

$$\|B_{kk}\|_{S_q} \geq b^{-\sigma} \kappa_{b-2,q} \rho_{b,q}(\sigma)^k. \quad (17)$$

Because $\kappa_{b-2,q} > 0$, the q th powers in equation (17) are not summable. By equation (16), $B_{b,\sigma} \notin S_q$.

When $\log_b \kappa_{b,q} > 1$, this argument treats the active digit-wall endpoint $\sigma = \log_b \kappa_{b,q}$ and the entire interval strictly below it but above the universal wall. When $\log_b \kappa_{b,q} \leq 1$, no $\sigma > 1$ satisfies $\rho_{b,q}(\sigma) \geq 1$, and proposition 5.2 already supplies necessity.

5.4 Binary paired-shell repair

At $b = 2$, the preceding pinch is illegal as a digit-wall witness because $B_{kk} = 0$ for every k ; see remark 4.2. We first remove the phase by equation (4) and work with the real matrix $B_{2,\sigma}$. Let

$$\hat{P}_j = P_{2j} + P_{2j+1}, \quad X_j := B_{2j+1,2j}.$$

The spaces $I_{2j} \oplus I_{2j+1}$ are mutually orthogonal. Since the two diagonal-shell blocks vanish and $B_{2,\sigma}$ is real symmetric,

$$\hat{P}_j B_{2,\sigma} \hat{P}_j = \begin{pmatrix} 0 & X_j^* \\ X_j & 0 \end{pmatrix}. \quad (18)$$

Its nonzero singular values are those of X_j , each repeated twice, so

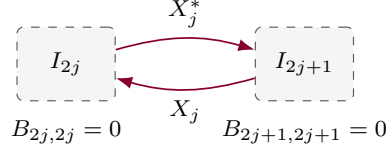
$$\|\hat{P}_j B_{2,\sigma} \hat{P}_j\|_{S_q}^q = 2 \|X_j\|_{S_q}^q. \quad (19)$$

Higher radices $b \geq 3$: same-shell pinch



$P_k B P_k = B_{kk}$, with
 $\|B_{kk}\|_{S_q} \gtrsim (\kappa_{b,q} b^{-\sigma})^k$.
 Orthogonal shell projections force non-summability when the ratio is at least one.

Binary radix $b = 2$: paired-shell pinch



$\widehat{P}_j B \widehat{P}_j = \begin{pmatrix} 0 & X_j \\ X_j^* & 0 \end{pmatrix}$, $\|\widehat{P}_j B \widehat{P}_j\|_{S_q}^q = 2\|X_j\|_{S_q}^q$,
 and $\|X_j\|_{S_q} \gtrsim (\kappa_{2,q} 2^{-\sigma})^{2j}$.

Figure 3: The two digit-wall pinching mechanisms after phase removal. For $b \geq 3$, nonzero same-shell blocks provide an orthogonal sequence. For $b = 2$, $X_j = B_{2j+1,2j}$ is the real weighted adjacent-shell block, and the mutually orthogonal two-shell compressions duplicate its singular values. Both lower bounds reject ratio 1; the right-hand construction is required because every binary same-shell block is zero.

Using equation (12) in the lower weighted comparison,

$$\|X_j\|_{S_q} \geq 2^{-3\sigma/2} (\kappa_{2,q} 2^{-\sigma})^{2j}. \quad (20)$$

Thus, throughout the full binary digit-norm nonmembership range $\kappa_{2,q} 2^{-\sigma} \geq 1$ with $\sigma > 1$, the q th powers of the paired compressions are nonsummable. This includes both the strict-below case, where the lower bound grows, and equality, where it does not decay. Equality is the exceptional repaired endpoint only because same-shell pinching is unavailable. As before, if the digit wall is at or below 1, the universal column wall is the active boundary.

5.5 Classification and operator classes

Theorem 5.3 (Sharp classification). *For every integer $b \geq 2$, every $1 \leq q < \infty$, and every $s \in \mathbb{C}$, with $\sigma = \Re s$,*

$$B_{b,s} \in S_q \iff \sigma > \max\{1, \log_b \kappa_{b,q}\}.$$

Proof. The forward implication fails at $\sigma \leq 1$ by proposition 5.2. For $\sigma > 1$ but $\kappa_{b,q} b^{-\sigma} \geq 1$, it fails by equation (17) when $b \geq 3$ and by equation (20) when $b = 2$. Two-sided unitary equivalence transfers these conclusions from real σ to complex s . The reverse implication is proposition 5.1. $\square$

Corollary 5.4 (Bounded, compact, Hilbert–Schmidt, and trace class). *The following are equivalent:*

$$B_{b,s} \text{ is bounded, } \quad B_{b,s} \text{ is compact, } \quad B_{b,s} \in S_2, \quad \sigma > 1.$$

Furthermore,

$$B_{b,s} \in S_1 \iff \sigma > \alpha_b = \log_b \tau_b > 1.$$

Proof. For $\sigma > 1$, corollary 3.2 gives $\log_b \kappa_{b,2} < 1$, so theorem 5.3 yields Hilbert–Schmidt, hence compact and bounded, membership. For $\sigma \leq 1$, boundedness fails by proposition 5.2. The trace-class statement is the case $q = 1$, together with $\tau_b > b$. $\square$

6 Traces, determinants, and temporal support

6.1 The diagonal trace

A diagonal entry is present precisely when a positive integer is carry-free with itself. Define

$$\mathcal{D}_b = \{m \geq 1 : \text{every base-}b \text{ digit of } m \text{ is at most } \lfloor \frac{b-1}{2} \rfloor\}. \quad (21)$$

The all-zero digit word would satisfy the digit restriction, but it is deleted because the source alphabet is $\mathbb{N}$.

Proposition 6.1 (Trace and its domain). *If $\sigma > \alpha_b$, then*

$$\text{Tr}(B_{b,s}) = \sum_{m \in \mathcal{D}_b} m^{-s}, \quad (22)$$

and the series is absolutely convergent. In this trace-class domain, the trace is zero for $b = 2$. For $b > 2$, it is strictly positive on the real half-line $s = \sigma > \alpha_b$.

Proof. By corollary 5.4, $B_{b,s}$ is trace class in the stated domain, so its trace is the absolutely convergent sum of its diagonal entries in the standard basis. The condition $m \perp_b m$ is exactly $2m_j < b$ at every digit, which is equation (21); the diagonal weight is m^{-s} . For $b = 2$, the only allowed digit is zero, so no positive vertex remains and $\mathcal{D}_2 = \emptyset$. For $b > 2$, the vertex 1 is in $\mathcal{D}_b$, and every term is positive when $s = \sigma$ is real. $\square$

The statement that the binary trace is zero in proposition 6.1 is an operator-trace statement only for $\sigma > \alpha_2$. Outside that domain, the diagonal or loop support is still empty, but we do not thereby define an operator trace. Likewise, positivity of the real Dirichlet series gives no zero-free statement for nonreal s .

6.2 Trace powers as absolutely convergent walk sums

Let a *based closed r -walk* be a tuple $(n_1, \dots, n_r) \in \mathbb{N}^r$ satisfying

$$n_i \perp_b n_{i+1}, \quad i \pmod{r},$$

where $n_{r+1} = n_1$. The base point matters, so cyclic rotations are not identified.

Proposition 6.2 (Trace-power expansion). *If $\sigma > 1$ and $r \geq 2$, then $B_{b,s}^r \in S_1$ and*

$$\text{Tr}(B_{b,s}^r) = \sum_{\substack{n_1, \dots, n_r \geq 1 \\ n_i \perp_b n_{i+1} \ (i \pmod{r})}} \prod_{i=1}^r n_i^{-s}. \quad (23)$$

The series on the right is absolutely convergent.

Proof. By corollary 5.4, $B_{b,s} \in S_2$ and is bounded. Hence $B_{b,s}^2 \in S_1$, and multiplying by bounded powers shows $B_{b,s}^r \in S_1$ for every $r \geq 2$.

Let $Q_K = \sum_{k=0}^K P_k$ and $T_K = Q_K B_{b,s} Q_K$. Because $B_{b,s}$ is Hilbert–Schmidt and Q_K increases strongly to the identity,

$$\|T_K - B_{b,s}\|_{S_2} \longrightarrow 0.$$

For $r = 2$, the ideal inequality gives

$$\|T_K^2 - B_{b,s}^2\|_{S_1} \leq (\|T_K\|_{S_2} + \|B_{b,s}\|_{S_2})\|T_K - B_{b,s}\|_{S_2}.$$

For $r > 2$, a telescoping expansion followed by bounded multiplication gives the same convergence in S_1 :

$$T_K^r \longrightarrow B_{b,s}^r \quad \text{in } S_1. \quad (24)$$

Thus the finite traces converge to the operator trace.

For each K , direct finite matrix multiplication yields

$$\mathrm{Tr}(T_K^r) = \sum_{\substack{n_1, \dots, n_r < b^{K+1} \\ n_i \perp_b n_{i+1} \ (i \bmod r)}} \prod_{i=1}^r n_i^{-s}.$$

To justify passage through the infinite sum, repeat the construction with the real matrix $B_{b,\sigma}$. Every term is nonnegative, the finite walk sums increase with K , and equation (24) shows that their traces converge to the finite number $\mathrm{Tr}(B_{b,\sigma}^r)$. Monotone convergence therefore gives

$$\sum_{\text{based closed } r\text{-walks}} \prod_{i=1}^r n_i^{-\sigma} < \infty.$$

This is exactly the sum of the absolute values of the complex- s terms. The complex finite sums may consequently pass term by term to the absolutely convergent limit, and their trace-norm limit identifies that limit with $\mathrm{Tr}(B_{b,s}^r)$. $\square$

6.3 Regularized and ordinary determinants

Corollary 6.3 (Determinant domains). *For $\sigma > 1$, the Hilbert–Schmidt regularized determinant*

$$\det_2(I - zB_{b,s})$$

is an entire function of z . On a sufficiently small neighborhood of $z = 0$, with the logarithm normalized to vanish at zero,

$$\log \det_2(I - zB_{b,s}) = - \sum_{r=2}^{\infty} \frac{z^r}{r} \mathrm{Tr}(B_{b,s}^r), \quad (25)$$

and each coefficient has the absolutely convergent walk expansion in equation (23). The ordinary trace and Fredholm determinant $\det(I - zB_{b,s})$ are asserted only for $\sigma > \alpha_b$.

Proof. For an S_2 operator, the regularized determinant $\det_2(I - zB) = \det((I - zB) \exp(zB))$ is entire in z ; this is the standard trace-ideal construction [9]. For, say, $|z| \|B_{b,s}\| < 1$, the norm-convergent logarithm and proposition 6.2 give equation (25). The ordinary Fredholm determinant requires $B_{b,s} \in S_1$, whose exact domain is $\sigma > \alpha_b$ by corollary 5.4. $\square$

We make no claim that equation (25) converges for all z , even though the determinant itself is entire in z . Nor do we construct a completed function of s , a functional equation, or a prescribed divisor.

6.4 Least periods of the graph shift

Let

$$\Sigma_b = \{(x_j)_{j \in \mathbb{N}_0} \in \mathbb{N}^{\mathbb{N}_0} : x_j \perp_b x_{j+1} \text{ for every } j \in \mathbb{N}_0\}$$

with the one-sided left shift. Repeating a cyclically admissible period- r word to the right gives a periodic point, and every periodic point arises this way; its least period is its least shift period. We discuss support only, independently of the complex weights.

Proposition 6.4 (Least-period support). *The least-period sets are*

$$\text{LPS}(\Sigma_2) = \{2, 3, \dots\}, \quad \text{LPS}(\Sigma_b) = \{1, 2, 3, \dots\} \quad (b > 2). \quad (26)$$

For every allowed length, there are infinitely many periodic points.

Proof. Distinct powers of b are pairwise carry-free because their nonzero digits occupy distinct positions. At $b = 2$, no positive vertex has a loop: any nonzero binary digit produces a carry when added to itself. Two distinct powers give a cyclic word of least period two. For each $r \geq 3$, a word of r distinct powers is cyclically admissible and cannot have a smaller period because all of its symbols are distinct.

For $b > 2$, the vertex 1 has a loop, providing least period one, and the same distinct-power construction gives every $r \geq 2$. Finally, shifting every chosen digit position upward by an arbitrary common amount produces infinitely many distinct witnesses of the same least period. $\square$

It follows that the unweighted fixed-point count is infinite at every allowed length (for $b = 2$, at every length at least two). Therefore the usual Artin–Mazur coefficients are not finite and no unweighted Artin–Mazur zeta function is claimed. The period statement is proved from support witnesses, not inferred from traces that could cancel for complex s .

7 Independent finite controls and proof audit

This section reports finite consistency evidence, not a proof of theorem 5.3. Every infinite statement above has an analytic proof independent of evaluator output, certificate status, or a finite cutoff.

Two separately constructed finite lanes were frozen. Lane A formed direct positive prefixes and checked carry-freeness by quotient and remainder. Lane B used digit automata and shell tensors. Each lane emitted 1,965 rows. Across the common corpus, 8,010 digit singular-value intervals overlapped, 420 direct weighted shell norms lay inside independently constructed endpoint envelopes, and the exact fields had no mismatch, missing row, extra row, or duplicate row.

An independently implemented proof audit checked four dependency and domain claims: the universal wall, the digit wall, the determinant domain, and the trace/least-period ledger. These checks do not replace the arguments in sections 5 and 6.

The hostile suite contained 39 atomic mutations. Across all consumers, it produced 68 designated rejections and 322 nondesignated acceptances, the latter checking that unrelated consumers did not reject an out-of-scope mutation. A further 76 physical or adversarial instances had zero survivors. Independent replays of both finite states passed in normal and adversarial import-path environments. The census is summarized in table 2.

The scope limitations are explicit. The source search was bounded; the validation set was finite; the Banach-ideal argument does not treat $0 < q < 1$; and the finite replay makes no archival, publication, or priority inference.

8 Discussion and conclusion

The carry-free operator separates cleanly into a local and a global mechanism. The local digit matrix supplies the exact growth factor $\kappa_{b,q}$; the global shell geometry supplies a second ratio $b^{(1-\sigma)/2}$. Both must be strictly smaller than one. A positive-density column makes the first global obstruction visible before any tensor calculation, whereas orthogonal pinching makes the digit obstruction sharp. In radix two, zero same-shell blocks change the pinching geometry

Finite/audit item	Count	Interpretation
Finite rows	1,965 per lane	two independently constructed lanes
Digit intervals	8,010	overlapping enclosures
Shell envelopes	420	direct values inside exact bounds
Exact-field discrepancies	0	finite consistency check
Atomic mutations	39	68 designated rejections
Nondesignated acceptances	322	expected consumer isolation
Physical/adversarial cases	76	normal and hostile replay inputs
Survivors	0	no detected hostile survivor

Table 2: Frozen finite and hostile-control census. A zero in the final row means that no listed hostile case survived its designated checks; it does not certify an infinite theorem.

but not the final formula: adjacent paired shells recover the entire relevant bad range and, in particular, its active equality endpoint.

The same ideal surface prevents domain slippage in the dynamical consequences. Trace powers and $\det_2$ begin at the Hilbert–Schmidt wall $\sigma > 1$; the ordinary trace and Fredholm determinant begin only at the trace-class wall $\sigma > \alpha_b$. Deleting the all-zero word then determines the loop and least-period ledgers. These support statements cannot be replaced by complex trace values, just as finite numerical agreement cannot replace the endpoint proofs.

The reusable lesson is methodological: an endpoint classification for an infinite digit operator must keep shell summability, column boundedness, and zero-block geometry separate. The present proof uses the triangle inequality in a Banach Schatten ideal, so it does not settle $0 < q < 1$. Other natural questions include quasi-Schatten thresholds and digit automata whose shell blocks are not pure tensor powers. We make no priority claim for the present combination, and no completed s -plane object, rational-prime primitive ledger, or spectral-divisor interpretation is asserted.

A Digit covariance details

We record the elementary inverse and boundary calculation behind proposition 3.1. Write $K_{ij} = \min(i, j)$ and let L have diagonal $2, \dots, 2, 1$ and first off-diagonals -1 . For an interior column $2 \leq j \leq b-1$,

$$(KL)_{ij} = -K_{i,j-1} + 2K_{ij} - K_{i,j+1} = \mathbf{1}_{\{i=j\}}.$$

The same direct difference at $j = 1$ uses $K_{i,0} = 0$, and the terminal column gives $-K_{i,b-1} + K_{i,b} = \mathbf{1}_{\{i=b\}}$. Thus $L = K^{-1}$.

For $Lv = \lambda v$, the interior recurrence and $v_0 = 0$ give

$$v_i = \sin(i\theta), \quad \lambda = 2 - 2\cos\theta.$$

The final equation $-v_{b-1} + v_b = \lambda v_b$ agrees with the interior recurrence exactly when $v_{b+1} = v_b$. Therefore

$$0 = \sin((b+1)\theta) - \sin(b\theta) = 2\cos\left(\frac{(2b+1)\theta}{2}\right)\sin\left(\frac{\theta}{2}\right).$$

The nonzero modes are $\theta_j = (2j-1)\pi/(2b+1)$, $1 \leq j \leq b$. Since the eigenvalues of K are $[4\sin^2(\theta_j/2)]^{-1}$, their positive square roots give equation (5).

Finally, column reversal changes C_b into a triangular matrix with ones on its diagonal. Hence

$$\prod_{j=1}^b s_j(C_b) = |\det C_b| = 1.$$

The strict inequality $\sum_j s_j(C_b) > b$ follows from strict arithmetic-geometric mean because $\sum_j s_j(C_b)^2 = b(b+1)/2 > b$, so the singular values cannot all equal one.

B Shell-factor bookkeeping

This appendix makes the coordinate deletion in proposition 4.1 explicit. Suppose $k > \ell$. Decompose the digits of $m \in I_k$ as

$$(m_k, \dots, m_{\ell+1}; m_\ell; m_{\ell-1}, \dots, m_0)$$

and those of $n \in I_\ell$ as

$$(n_\ell; n_{\ell-1}, \dots, n_0).$$

The first group of digits of m has $(b-1)b^{k-\ell-1}$ possibilities and is automatically compatible with the zero digits of n . It is therefore a row-repetition coordinate. At position ℓ , the column digit n_ℓ is nonzero. The row $m_\ell = b-1$ has no compatible column and is deleted. The relabeling

$$a = m_\ell \in \{0, \dots, b-2\}, \quad c = n_\ell - 1 \in \{0, \dots, b-2\}$$

turns $m_\ell + n_\ell < b$ into $a + c < b-1$, which is C_{b-1} . Each lower position contributes C_b . This proves equation (9); no all-zero positive vertex has been added.

If $k = \ell$, both leading digits are nonzero. Any compatible pair lies in $\{1, \dots, b-2\}^2$, and subtracting one from both digits changes

$$m_k + n_k < b \quad \text{to} \quad (m_k - 1) + (n_k - 1) < b - 2.$$

This is C_{b-2} ; the lower positions again contribute $C_b^{\otimes k}$. For $b = 2$ the compatible leading-digit set is empty, giving the zero block C_0 . The $k < \ell$ formula is the transpose of the $k > \ell$ formula because the carry-free relation is symmetric.

The weighted comparison is also insensitive to zero rows and columns. For the lower estimate, the exact identity

$$A_{k\ell} = D_k^{-1} B_{k\ell} D_\ell^{-1}$$

and the shellwise bounds on the inverse diagonals lose only the fixed factor b^σ . This uniformity, rather than the exact value of the factor, is what allows the lower bounds to remain nondecaying at equality.

C Endpoint pinching and determinant convergence

C.1 Direct-sum accounting

For a finite family of mutually orthogonal projections $Q_1, \dots, Q_N$, the block-diagonal map

$$T \mapsto \sum_{j=1}^N Q_j T Q_j$$

is an average over diagonal sign unitaries and is therefore contractive on every S_q , $1 \leq q < \infty$. Letting $N \rightarrow \infty$ gives the countable version used in equation (16). Because the image is a direct sum,

$$\left\| \bigoplus_j Q_j T Q_j \right\|_{S_q}^q = \sum_j \|Q_j T Q_j\|_{S_q}^q.$$

For $b \geq 3$, take $Q_j = P_j$ and use equation (17). For $b = 2$, take $Q_j = \hat{P}_j = P_{2j} + P_{2j+1}$. If

$$H_j = \begin{pmatrix} 0 & X_j^* \\ X_j & 0 \end{pmatrix},$$

then

$$H_j^* H_j = \begin{pmatrix} X_j^* X_j & 0 \\ 0 & X_j X_j^* \end{pmatrix}.$$

The nonzero eigenvalues of the two diagonal blocks agree, proving $\|H_j\|_{S_q}^q = 2\|X_j\|_{S_q}^q$. Together with equation (20), this rejects both $\kappa_{2,q}2^{-\sigma} > 1$ and equality whenever the digit wall is relevant above the universal wall.

C.2 Finite-shell passage for trace powers

Let $T = B_{b,s} \in S_2$ and $T_K = Q_K T Q_K$. The elementary Hilbert–Schmidt approximation

$$\|T - Q_K T Q_K\|_{S_2} \leq \|(I - Q_K)T\|_{S_2} + \|Q_K T(I - Q_K)\|_{S_2} \longrightarrow 0$$

justifies the first limit in the proof of proposition 6.2. The telescoping identity

$$T_K^r - T^r = \sum_{j=0}^{r-1} T_K^j (T_K - T) T^{r-1-j}$$

is estimated in S_1 by placing the S_2 difference next to one S_2 factor and bounding the remaining factors in operator norm. For the endpoint terms one first uses the $r = 2$ estimate and then bounded multiplication. Hence $T_K^r \rightarrow T^r$ in trace norm for every $r \geq 2$.

For real σ , the finite closed-walk sums have nonnegative terms and increase with K . Their bounded trace limit supplies an integrable majorant for the complex- s sums. This is the required exchange of the finite-shell limit with the walk expansion; trace-class membership alone is not being used as an unexplained termwise interchange.

Finally, for $|z|\|T\| < 1$, the scalar identity

$$\log(1 - z\lambda) + z\lambda = - \sum_{r \geq 2} \frac{z^r \lambda^r}{r}$$

may be summed over the eigenvalues using the S_2 bound. This proves the local series in equation (25). Analytic continuation defines the entire determinant, not a global choice of its logarithm or global convergence of that series.

D Reproducibility and evidence ledger

The data-driven assets are generated by `scripts/generate_paper_assets.py` from the content-addressed `figures/data/canonical_summary.json`. The generator recomputes the critical curves from equation (6), checks the summary’s finite bindings and evidence ceiling, emits the tables, and records input/output SHA-256 values in `figures/data/ASSET_DIGESTS.json`. Thus the displayed decimals and census values do not depend on manual transcription.

Before asset generation, the canonical extraction checked the finite contract, two independently produced output trees, result ledgers, exact cross-lane fields, interval overlap, and mutation and adversarial counts. A separate replay then checked both finite states in normal and adversarial import environments. The figure and tables are deterministic evaluations of the exact digit formula or literal reports of that frozen census.

Evidence types remain disjoint:

- the arguments in sections 3 to 6 are the mathematical proof;
- four independent audit checks examine proof dependencies and strict domains;
- finite tensors, intervals, cutoffs, mutations, and adversarial replays are falsification and implementation controls only; and
- the directed literature search is bounded and cannot establish priority.

No external archival or publication-attestation artifact is claimed in this manuscript.

References

- [1] J. Alman, Y. Guan, and A. Padaki. Smaller low-depth circuits for kronecker powers. In *Proceedings of the 2023 Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 4159–4187. Society for Industrial and Applied Mathematics, 2023.
- [2] R. Bacher and R. Chapman. Symmetric pascal matrices modulo p . *European Journal of Combinatorics*, 25(4):459–473, 2004.
- [3] D. Chistikov, S. Iván, A. Lubiw, and J. Shallit. Fractional coverings, greedy coverings, and rectifier networks. In *34th Symposium on Theoretical Aspects of Computer Science (STACS 2017)*, volume 66 of *Leibniz International Proceedings in Informatics*, pages 23:1–23:14. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2017.
- [4] P. R. Christopher and J. W. Kennedy. Binomial graphs and their spectra. *The Fibonacci Quarterly*, 35(1):48–53, 1997.
- [5] E. E. Kummer. Über die ergänzungssätze zu den allgemeinen reciprocitätsgesetzen. *Journal für die reine und angewandte Mathematik*, 1852(44):93–146, 1852.
- [6] J. D. LaGrange. Eigenvalues of boolean graphs and pascal-type matrices. *International Electronic Journal of Algebra*, 13:109–119, 2013.
- [7] N. Linial and A. Shraibman. Lower bounds in communication complexity based on factorization norms. In *Proceedings of the Thirty-Ninth Annual ACM Symposium on Theory of Computing*, pages 699–708. ACM, 2007.
- [8] É. Lucas. Sur les congruences des nombres eulériens et des coefficients différentiels des fonctions trigonométriques suivant un module premier. *Bulletin de la Société Mathématique de France*, 6:49–54, 1878.
- [9] B. Simon. *Trace Ideals and Their Applications*, volume 120 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2 edition, 2005.